3Laws

Release beta

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Chapter 1

Introduction

Here is the public documentation for 3LawsRobotics Products:

- Robot Diagnostic Module (Robot Diagnostic Module):

Chapter 2

Robot Diagnostic Module

The **Robot Diagnostic Module (RDM) of 3LawsRobotics** is a powerful tool to gather analytics about your robot. It provides low level metrics such as CPU usage, signal delays but also high level safety critical metrics like safety envelope violation, planning inconsistency or sensor damage detection.

All of these metrics can be viewed on your web browser at 3laws.app.

In order to get your account setup and use freely this RDM, please contact contact@3lawsrobotics.com

This product is still at a Beta development stage. New features are developed every month and some breaking changes can still happened. The 3LawsRobotics Team is available to answer your question.

For now only a ROS1, ROS2 interface is available but we intend to extend the offer of possible interfaces in the near future with support for MavLink or MQTT for instance.

2.1 Installation

The Robot Diagnostic Module is distributed as a debian package. A package is available for each of the following configuration

Ubuntu Distribution	ROS1 version	ROS2 version	Architecture
22.04	N/A	Humble/Iron	amd64/arm64v8
20.04	Noetic	Galactic/Foxy	amd64/arm64v8
18.04	Melodic	N/A	amd64/arm64v8

Here is the one liner that will download a public script that will, based on your system download the right package.

```
sudo bash -c "$(wget -qO - https://raw.githubusercontent.com/3LawsRobotics/3laws/master/rdm/install.sh)"
```

If you want to install a specific package in a non interactive mode, here is the process:

- Download the script:
  `wget https://raw.githubusercontent.com/3LawsRobotics/3laws/master/rdm/install.sh`
- Make it executable
  `chmod +x install.sh`
- Run it with your arguments
  `sudo ./install.sh [-hyf] [-r <ROS_DISTRO>] [-a <ARCH>] [-v <UBUNTU_VERSION>]` if `-yf -r <ROS_DISTRO> -a <ARCH> -v <UBUNTU_VERSION>` specified, the script is non interactive

2.2 Launch the RDM

To launch the RDM, you only have to source your ros distribution: `source /opt/ros/<DISTRO>/setup.sh` and to run the command `ros2 launch lll_rdm rdm.launch.py`

You can add the following command line options:

- use_sim_time: <Boolean, default: false> Specify if the RDM use the ros time or the machine time
- config_filename:
  <String> Path to the config file if not in the `/opt/3lawsRoboticsInc/robot_diagnostic_module/config` folder (**needs to start with /**) or just the config file name inside the default folder.
- robot_id: <String> name of your robot
- log_level: <trace | debug | info | error, default: info> Log level of the RDM
- dry_run: <Boolean> Start the RDM, check the config file and turn off without error if everything has been launched correctly

2.3 Configuration

The configuration is written inside a YAML. file. This file can be stored anywhere but the default folder is `/opt/3lawsRoboticsInc/robot_diagnostic_module/config`

Other path can be specified in the command line launch command

The Robot Diagnostic Module is shipped with a configuration template that can be found here
`/opt/3lawsRoboticsInc/robot_diagnostic_module/config/template_config.yaml`

In order to have the more user friendly configuration experience, a lot of the fields are optional. The configuration can be build alongside your robot so that it does not represent a burden for the developer.

2.3.1 Credentials

Credentials are used to connect to the databases to store the data. Here is the structure and the field detail.

```
credentials:
  company_id: Company name
  robot_id: Robot Name
  influx_credentials:
    database_token: INFLUX token provided by 3lawsRobotics
  clickhouse_credentials:
    username: Company name
    password: CLICKHOUSE password provided by 3lawsRobotics
```

2.3.2 Diagnostic Configuration

This part of the configuration is here to define diagnostic specific options.

Legend:

- <Distance array> An array of distance in meter
- <Percentage array> An array of percentages (between 0 and 100)
- <Time array> An array of time. Allowed unit: s, ms, min

- <Ratio Array> An array of ratio (between 0 and 1). Used when values are normalized
- <Severity> An enum: ok | minor | severe | critical | ~

If a value is set to [] or ~ the RDM will not flag event for this specific case. In the case of an array option, the first value is the threshold to detect a critical event, the second for a severe event and so on. So for example, a Distance Array like this [0.001, 0.005, 0.01] will flag a minor event if the value goes under 0.01, then reevaluates it to 0.005 if the value is inferior to 0.005 and finally raise the severity to critical if the value goes under 0.001

```
diagnostic_config:
  timeout_factor: <Int> Specify how much time the max signal rate has to be wait
                        to declare a topic timed out. This trigger an event.
  max_signals_delay: <Time <int> ms/s/min> Specify the maximum delay between
                                           the send time and the received time
  incident_detection: This structure specify the severity and the threshold for
the event metrics
    min_event_time: <Time <int> ms/s/min> Minimal time to flag an event
    min_displayed_severity: <Enum: ok|minor|severe|critical> Minimal criticality
to flag an event
    collision:
      limit_dists: <Distance array>
    computer:
      limit_cpu_loads: <Percentage array>
      limit_disk_usages: <Percentage array>
      limit_ram_usages: <Percentage array>
    clock:
      limit_utc_deviations: <Time array>
      limit_rtc_deviations: <Time array>
    node_health:
      timeout_severity: <Severity>
      not_ok_severity: <Severity>
      limit_delay_bounds: <Ratio Array>
      limit_rate_bounds: <Ratio Array>
    obstruction:
      obstruction_severity: <Severity>
    signal_health:
      nan_severity: <Severity>
      zero_severity: <Severity>
      subnormal_severity: <Severity>
      inf_severity: <Severity>
      bad_timestamp_severity: <Severity>
      limit_bounds: <Ratio Array>
    dynamic:
      limit_xdot_difference: <Ratio Array>
      input_domain_severity: <Severity>
      state_domain_severity: <Severity>
    tracking:
      limit_tracking_difference: <Ratio Array>
  upload: This configuration allow the user to reduce the bandwidth
          by not uploading high frequency metrics
    high_frequency:
      state: <Boolean>
      input: <Boolean>
      path_tracking_error: <Boolean>
      control_tracking_error: <Boolean>
```

2.3.3 Interface Configuration

The interface structure is here to get precision on the interface. Right know only ROS interfaces are available, the interface option are here to specify Quality of Service and topic types

```
interface:
```

```
  retimestamp: never/always/if_zero specify if the timestamp has to be set

  ros_topics_info:
    /example_topic!:
      type: sensor_msgs/LaserScan
    /example_topic2:
      type: lll_msgs/Float64VectorStamped
    /example_topic3:
      type: nav_msgs/Odometry
```

Supported topics type are all ROS default types and 3lawsRobotics custom types (ns: lll_msg, lll_rdm_msg)

2.3.4 Robot Description

This part is the most important one. The more accurate the information provided here, the more insightful the RDM will be.

Here is a template of this section. Each sub structure is optional but once a substructure is declared, all its fields must be filled (unless Optional is specified)

```
robot_description:
  # robot_type and associated model_type options are (`*` not available yet):
  #   mobile_robot -> differential_drive, *omni_directional, *ackerman, bicycle
  #   legged_robot -> *bipedal, *quadruped
  #   land_vehicle -> *basic
  #   aerial_vehicle -> *fixed_wing, *helicopter, *multi_rotor
  #   surface_vehicle -> *sail_boat, *outboard_boat, *inboard_boat, *jet_boat
  #   underwater_vehicle -> *omni_directional
  #   spacecraft -> *omni_directional
  #   manipulator -> *fixed_base
  #   unclassified -> *unknown, *first_order
  robot_type: mobile_robot

  kinematic_model: # Kinematic model of the robot
    base_frame_id: robot # Name of the base frame of the robot
    geometry: # Geometry of the robot
      shape_type: box # Type of the geometry shape (should be one of [point, box,
sphere, ellipsoid, capsule, cone, cylinder, mesh])
      shape_params: # Parameters for the chosen shape_type (See bottom of this
file for more info)
        x_size: 0.5
        y_size: 0.4
        z_size: 0.3
      pose: # [optional] Pose of shape w.r.t base frame
        translation: [0., 0., 0.] # [optional] Translation w.r.t base frame
[x,y,z], default [0,0,0]
        rotation: [1., 0., 0., 0.] # [optional] Rotation quaternion w.r.t base
frame [w,x,y,z], default [1,0,0,0]

  dynamical_model: # [optional] Behavior model of the robot
    model_type: bicycle # Type of dynamical model for selected robot_type
    state_domain: # Region of the state space in which the dynamical model has
been validated
      # Defined as a a convex polytope S={x in Rn | lb <= M.x <= ub}
      ub:
        [1., 2., 3.] # If simple==true, must have same size as state of
model_type,
        # otherwise must have size n_cstr
      lb:
        [-1., -2., -3.] # If simple==true, must have same size as state of
model_type,
        # otherwise must have size n_cstr
      simple: false # [optional], If true, assumes M is an identity matrix
```

```
(default: false)
      n_cstr: 3 # [optional if simple==true] Number of rows in M
      M: # [optional if simple==true] Either a single sequence representing
the diagonal of M, or a sequence of the rows of M
        - [1., 0., 0., 0., 0.]
        - [0., 1., 0., 0., 0.]
        - [0., 0., 1., 0., 0.]
    input_domain: # Region of the input space accessible for control purposes and
in which the dynamical model has been validated
      # Same representation as 'state_domain'
      ub:
        [1., 2.] # If simple==true, must have same size as input of model_type,
        # otherwise must have size n_cstr
      lb:
        [-1., -2.] # If simple==true, must have same size as input of model_type,
        # otherwise must have size n_cstr
      simple: true
    process_noise_covariance:
      [1., 1., 1., 1., 1.] # Either a single sequence representing the diagonal of
the process noise covariance matrix,
      # or a sequence of its rows. Must have same size as state of model_type
    model_param: # [variant] Parameters for the chosen model_type (See bottom of
this file for more info)
      wheel_dx: 1.
      origin_dx: 1.
      tau_vel: 1.
      tau_steer: 1.

  mission_manager: # [optional]
    extra_topics: ~ # [optional] (See bottom of this file for more info)
    process_name: ~ # [optional] (See bottom of this file for more info)
    finite_states: # Finite states of the robot
      - interface_id:
          /status # Name of the ros topic.
          # Supported types: [(default) std_msgs/String, std_msgs/FloatXX,
std_msgs/Bool, std_msgs/Char, std_msgs/Byte, std_msgs/IntXX, std_msgs/UIntXX]
        sender_id: state_machine # Display name for sender of this state
        state_id: status # Identifier for this state, "<sender_id>.<state_id>"
must form a UNIQUE identifier among all signals
        signal_min_rate: 1s
      - interface_id: /search_mode
        sender_id: state_machine
        state_id: search_mode
        signal_min_rate: 1s

  sensors: # [optional]
    extra_topics: ~ # [optional] (See bottom of this file for more info)
    process_name: ~ # [optional] (See bottom of this file for more info)
    batteries: [] # Coming soon!
    cameras: [] # Coming soon!
    gps: [] # Coming soon!
    imus: [] # Coming soon!
    laserscans: # Planar laser scanners
      - interface_id:
          /laserscan_1_topic # Name of the ros topic.
          # Supported types: [(default) sensor_msgs/LaserScan]
        sensor_id: laserscan_1 # Display name for this laserscan, must be UNIQUE
among all laserscans
        signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
        specs:
          n_rays: 1000 # Expected number of rays in the laserscan
          angle_min: -3.14 # Minimum ray angle
          angle_max: 3.14 # Maximum ray angle
```

```
          range_min: 0. # Minimum ray range
          range_max: 1000. # Maximum ray range
          noise_one_sigma: 0.025 # Expected standard_error of the sensor (given by
the manufacturer, often like: precision = +-2sigma)
        transform: # Specification of frame w.r.t which the measurement is
expressed
          parent_frame_id: robot # Id of parent frame
          pose: # [optional] Pose w.r.t parent frame
            translation: [0., 0., 0.] # [optional] Translation w.r.t parent frame
[x,y,z], default [0,0,0]
            rotation: [1., 0., 0., 0.] # [optional] Rotation quaternion w.r.t
parent frame [w,x,y,z], default [1,0,0,0]
    lidars: [] # Coming soon!
    loadcells: # Force and torque measurement sensor, 6 axis by default
      - interface_id:
          /end_effector_wrench # Name of the ros topic.
          # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
        sensor_id: end_effector_loadcell # Display name for this loadcell, must be
UNIQUE among all loadcells
        signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
        transform: # Specification of frame w.r.t which the measurement is
expressed
          parent_frame_id: robot # Id of parent frame
          pose: # [optional] Pose w.r.t parent frame
            translation: [0., 0., 0.] # [optional] Translation w.r.t parent frame
[x,y,z], default [0,0,0]
            rotation: [1., 0., 0., 0.] # [optional] Rotation quaternion w.r.t
parent frame [w,x,y,z], default [1,0,0,0]
        # axis_mask: # [optional] Define which of the 6 force/torque axes in SE3
the loadcell signals correspond to: [Fx, Fy, Fx, Mx, My, Mz].
        #   # If not specified or null, assumes all 6 axes.
        #   # Cannot be empty or longer than 6. Index must be between 0 and 5
included.
        #   [0, 5] # Corresponds to a 2 axis loadcell [Fx,Mz]
        noise_one_sigma: [1., 1., 1., 1., 1., 1.] # Noise characteristics of
loadcell axes. Must have same size as axis_mask
        bounds: ~ # [optional] (See bottom of this file for more info)

  perception: # [optional]
    obstacles: # [optional] List of obstacles
      interface_id: /obstacles # Name of the ros topic. # Supported types:
[(default) lll_msgs/ObjectArray]
      signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
      meshes: # List of meshes to be loaded
        []
        # - id: sphere # Mesh identifier, must be UNIQUE among all meshes
        #   data: # Mesh data
        #     mesh_file: sphere.stl # Path to mesh file
        #     mesh_type: stl # Type of mesh file
        #     mesh_units: mm # Unit of mesh file

  localization: # [optional]
    extra_topics: ~ # [optional] (See bottom of this file for more info)
    process_name: ~ # [optional] (See bottom of this file for more info)
    state_estimation: # [optional]
      interface_id:
        /state # Name of the ros topic.
        # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
      signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
```

```
      state_size: 5 # Size of the state vector
      # mask:
      #   [0, 1, 2, 3, 5] # [optional] If only a subset of the vectorized message
actually constitute the state vector
      #   # use this mask to extract the relevant data : state[i] =
msg_vectorized[mask[i]].
      #   # Must be of size 'state_size', and not contain indices greater than
the size of vectorized message.
      #   # If not specified or null, will be [0, ..., state_size-1]
      bounds: ~ # [optional] (See bottom of this file for more info)

    odometry:
      - interface_id:
          /odom_node_0_topic # Name of the ros topic.
          # Supported types: [(default) nav_msgs/Odometry]
        odom_id: odom_node_0 # Display name for this odometry source, must be
UNIQUE among all odometry
        signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
        se2_only: true # [optional] Consider only SE2 projection of pose and twist
(default: false)
        position_bounds: # [optional] Position part of the odometry. Components
are [x,y,z] or [x,y] if se2_only==true (See bottom of this file for more info)
          norm_type: none
          norm_upper_bound: 1.
          norm_lower_bound: 0.
          upper_bounds: [1., 1.]
          lower_bounds: [-1., -1.]
          rates_upper_bounds: []
          rates_lower_bounds: []
        orientation_bounds: ~ # [optional] Same fields as position. Components are
[roll,pitch,yaw] or [yaw] if se2_only==true (See bottom of this file for more
info)
        velocity_linear_bounds: ~ # [optional] Same fields as position. Components
are [vx,vy,vz] or [vz,vy] if se2_only==true (See bottom of this file for more
info)
        velocity_angular_bounds: ~ # [optional] Same fields as position.
Components are [wx,wy,wz] or [wz] if se2_only==true (See bottom of this file for
more info)

  path_planning: # [optional]
    extra_topics: ~ # [optional] (See bottom of this file for more info)
    process_name: ~ # [optional] (See bottom of this file for more info)
    paths:
      - interface_id:
          /desired_path # Name of the ros topic.
          # Supported types: [(default) lll_msgs/Trajectory, nav_msgs/Path,
trajectory_msgs/JointTrajectory]
        path_id: main_path # Display name for this path, must be UNIQUE among all
paths
        trajectory_state_size: 7 # Size of the trajectory state vector
        signal_min_rate: 1min # Maximum time without receiving data before signal
is considered timed out
        # state_mask:
        #   [0, 1, 2, 3, 4, 5, 6] # [optional] If the path only corresponds to
a subset of the state_estimation vector,
        #   # use this mask to extract the relevant data : trajectory_state[i] =
state_estimation[state_mask[i]].
        #   # Must be of size 'trajectory_state_size', and not contain indices
greater than state_estimation.state_size.
        #   # If not specified or null, will be [0, ..., trajectory_state_size-1]
        tracking_error_bounds: ~ # [optional] Bounds on controller's tracking
error : path_state - actual_state (See bottom of this file for more info)
```

```
  control: # [optional]
    extra_topics: ~ # [optional] (See bottom of this file for more info)
    process_name: ~ # [optional] (See bottom of this file for more info)
    setpoint_tacking_controllers: # PID like controllers
      - controller_id: velocity_controller # Display name for this controller,
must be UNIQUE among all controllers
        state_size: 1 # Size of controller setpoint
        input_size: 1 # Size of control input computed by controller
        desired_state:
          interface_id: /controller_cmd_topic # Name of the desired state ros
topic.
          # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
          mask: [0] # [optional] If only a subset of desired_state_topic_id vector
is actually used by controller,
          # use this mask to extract the relevant data : desired_state_used[i] =
desired_state_received[desired_state_mask[i]]
          signal_min_rate: 1s # Maximum time without receiving data before signal
is considered timed out
          bounds: ~ # [optional] Bounds on desired state (See bottom of this file
for more info)

        actual_state:
          interface_id: /controller_state_topic # Name of the actual state ros
topic.
          # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
          mask: ~ # [optional] Same as desired_state_mask
          signal_min_rate: 1s # Maximum time without receiving data before signal
is considered timed out
          bounds: ~ # [optional] Bounds on actual state (See bottom of this file
for more info)

        control_input:
          interface_id: /controller_input_topic # Name of the control input ros
topic.
          # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
          mask: ~ # [optional] Same as desired_state_mask
          signal_min_rate: 1s # Maximum time without receiving data before signal
is considered timed out
          bounds: # [optional] Bounds on controller's control input (See bottom of
this file for more info)
            norm_type: none
            norm_upper_bound: 1.
            norm_lower_bound: 0.
            upper_bounds: []
            lower_bounds: []
            rates_upper_bounds: [1.]
            rates_lower_bounds: [-1.]
        tracking_error_bounds: # [optional] Bounds on controller's tracking error
: desired_state - actual_state (See bottom of this file for more info)
          norm_type: none
          norm_upper_bound: 1.
          norm_lower_bound: 0.
          upper_bounds: [1.]
          lower_bounds: [-1.]
          rates_upper_bounds: []
          rates_lower_bounds: []

    actuators: # Robot actuation
      combined: # Combined actuation vector
        interface_id:
          /control_input # Name of the ros topic publishing the complete robot
```

```
actuation vector.
          # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
        input_size: 3 # Size of the combined input vector
        signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
        bounds: ~ # [optional] (See bottom of this file for more info)
        # mask:
        #   [0,1,3] # [optional] If only a subset of the vectorized message
actually constitute the combined input vector
        #   # use this mask to extract the relevant data : input[i] =
msg_vectorized[mask[i]].
        #   # Must be of size 'input_size', and not contain indices greater than
the size of the vectorized message.
        #   # If not specified or null, will be [0, ..., input_size-1]

    supervisors: # 3Laws AI Supervisors
      - interface_id: /main_supervisor_topic # Name of the supervisor data ros
topic.
        supervisor_id: main_supervisor # Display name for this supervisor, must be
UNIQUE among all supervisor
        signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out

  extras: # [optional]
    passthrough_metrics: # Generic passthrough for scalar metric signals
      - interface_id:
          /metric_1_topic # Name of the ros topic.
          # Supported types: [(default) std_msgs/Float64, std_msgs/Float32,
std_msgs/Bool, std_msgs/Char, std_msgs/Byte, std_msgs/IntXX, std_msgs/UIntXX]
        metric_id: metric_1 # Display name for this metric, must be UNIQUE among
all passthrough metrics
        metric_group_id:
          position # [optional] Group this signal belongs to.
          # Metrics of the same group are plotted on the same graph in 3laws.app

    clocks:
      - interface_id:
          /custom_clock # Name of the ros topic.
          # Supported types: [(default) rosgraph_msgs/Clock]
        clock_id: my_clock # Display name for this clock, must be UNIQUE among all
clocks
        signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out

    signals: # Generic floating point multidimensional signal values sanity and
bounds checking
      - interface_id:
          /test_signal_topic # Name of the ros topic.
          # Supported types: [(default) lll_msgs/Float64VectorStamped, any other
vectorizable type (see bottom of this file)]
        sender_id: test_signal_node # Display name of sender node
        signal_id: test_signal # Display name of this signal,
"<sender_id>.<signal_id>" must form a UNIQUE identifier among all signals
        signal_size: 1 # Size of this signal
        signal_min_rate: 1s # Maximum time without receiving data before signal is
considered timed out
        # mask:
        #   [2] # [optional] If only a subset of vectorized message actually
constitute the signal vector
        #   # use this mask to extract the relevant data : signal[i] =
msg_vectorized[mask[i]].
        #   # Must be of size 'signal_size', and not contain indices greater than
the size of the vectorized message.
```

```
        #   # If not specified or null, will be [0, ..., signal_size-1]
        bounds: ~ # [optional] (See bottom of this file for more info)

    nodes: # Generic node health checking metric
      - node_id: test_node # Display name of node, must be UNIQUE among all nodes
        # text_log_interface_id:
        #   /test_node_log # [optional] Name of the ros topic publishing log info
for that node.
        #   # Supported types: [(default) rcl_interfaces/Log]
        # process_name: # [optional] (See bottom of this file for more info)
        #   test_node_exec.
        topics: # List of topics published by the node (only available in ros2
humble and up)
          - interface_id:
              /test_node_topic_1 # Name of the ros topic. Associated
'interface.message_type_map.<interface_id>' must be specified.
              # Supported types: [builtin_interfaces/*, geometry_msgs/*,
lll_msgs/*, nav_msgs/*, rcl_interfaces/*, rosgraph_msgs/*, sensor_msgs/*,
std_msgs/*, trajectory_msgs/*, visualization_msgs/*]
            topic_id: test_node_topic_1 # Display name for this topic, must be
UNIQUE among all topics of each node
            signal_min_rate: 1s # Maximum allowed duration without receiving data
      - node_id: rosout
        text_log_interface_id: /rosout # If equal to '/rosout', uses 'name' field
of incoming rcl_interfaces/Log message as node_id for text_log message
        topics: []
```

2.3.5 Scheme

The configuration has **4** distinct parts

- Credentials (Credentials):
  This part is here to specify the credentials to access the databases. These credentials are provided to you by email once your 3LawsRobotics account is created
- Diagnostic Configuration (Diagnostic Configuration):
  This part is the more important one. It describe your robot desired behavior, the interface data stream access for the RDM to monitor.
- Interface Configuration (Interface):
  Here are specified the diagnostic specific options such has the criticality of a certain event or the threshold above which a incident is flagged
- Robot Description (Robot Description):
  And this last part allow the user to specify interface specific options such has ROS Quality of Service